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Optimal Design of Coastal Defences

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Abstract

The use of coastal defences is essential to protect hinterland communities from flooding and erosion risk. This study investigated the hydrodynamic performance of hard coastal defences using the Smoothed Particle Hydrodynamics (SPH) method implemented in DualSPHysics, with the aim of identifying an optimal design configuration. A vertical seawall case was reproduced and benchmarked against published SPH and analytical solutions, achieving an average error of less than 2% and instilling confidence of the numerical setup. Four coastal defence geometries were analysed in a two-dimensional wave flume namely a vertical seawall, a sloped impermeable revetment and two hybrid structures which combined both revetment and seawall with varying height elevations. Results revealed clear trade-offs between overtopping performance and horizontal loading. The seawall recorded minimal overtopping volume but experienced high horizontal loading. The revetment on the other hand, it reduced the maximum horizontal loading but induced excessive overtopping volumes. The hybrid structures combined both advantages, with revetment crest and parapet height identified as critical parameters. Raising the revetment crest above Still Water Level (SWL) effectively minimised the parapet loading but increased overtopping. Therefore, an elevated revetment must be paired with a taller parapet to achieve optimal performance. Comparison with the EurOtop overtopping limits showed only the vertical seawall and hybrid with raised parapet height maintained acceptable discharge rates under breaking wave conditions. These findings demonstrated the capability of SPH for coastal defence design and suggested that a hybrid defence structure can strike a balance between structural stability and overtopping resilience.

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1 Introduction

1.1 Background

Coastlines represent the dynamic boundary between marine and terrestrial environments, the place where the land meets the sea. This boundary is shaped by complex and continuous processes such as wave action, tidal flows, sediment transport, and changes in sea level, contributing to patterns of erosion and accretion over time. These natural processes can pose a significant threat to communities and infrastructure located near coastal zones through land loss and increased flood risk. For example, the UK Met Office predicts that sea levels will rise by 1.1 m by 2100 under high-emission scenarios (Office 2023). Moreover, human populations have historically settled in coastal areas, and today approximately 10% of the global population resides within 5 kilometres of the coast (Cosby 2024). This is mainly due to the economic importance of coastal cities, which serve as hubs for trade, transportation, and employment.

In response, coastal defence systems have been developed to mitigate these hazards by resisting shoreline retreat and reducing flooding risk. Effective and sustainable coastal defence systems are therefore essential to ensure long-term safety and resilience of coastal populations. Coastal defence systems can broadly be categorised into hard engineering structures such as seawalls, revetments, groynes and breakwaters and soft engineering measures which include beach nourishment, dune restoration, saltmarsh vegetation and managed realignment (French 2001). Hard engineering surfaces provide immediate protection by acting as a physical barrier to wave action but are typically expensive, inflexible and environmentally unsustainable. In contrast, soft engineering approaches support natural processes and ecological resilience but may provide less protection under extreme events. Hybrid strategies that combine both hard and soft approaches are promoted as a sustainable long-term solution to coastal defences. The design of hard coastal structures requires careful consideration prior to construction, and it is critical to balance the hydrodynamic loading and overtopping performance. Excessive horizontal forces can compromise structural stability, while overtopping can lead to flooding, damaging communities and infrastructure in the hinterland. Consequently, both force and overtopping discharge are key criteria in coastal

engineering design and are addressed explicitly in the design guidance such as the EurOtop Manual (Van der Meer 2018).

Therefore, analysing the performance of various coastal defences is essential before the implementation of a coastal defence structure. This can be achieved through physical experiments utilising laboratory wave flumes, which provide accurate data under controlled conditions. However, physical modelling is often costly, time-consuming, and can be constrained by the limited configurations that it can be applied to. The rapid advancement of high-performance computing has enabled the adoption of numerical modelling, allowing efficient simulation of a wide range of configurations and wave conditions.

Numerical modelling approaches can be divided into mesh-based methods and meshless methods. Mesh-based approaches, like Reynolds-Averaged Navier-Stokes solvers, are widely used but often require complex meshing and struggle with highly non-linear processes such as wave breaking and overtopping. In contrast, the latter, Smoothed Particle Hydrodynamics (SPH), has gained significant traction for coastal engineering applications due to its ability to simulate highly non-linear free-surface flows and complex wave-structure interactions (Altomare 2015). SPH discretises the fluid domain into multiple particles that move with the flow, eliminating mesh generation issues and naturally capturing processes such as run-up and overtopping. The open-source implementation of SPH, DualSPHysics, has been widely applied and validated for studying free surface flows (Monaghan 1992), (Altomare 2015).

This study applied DualSPHysics to simulate wave-structure interactions for a set of representative 2D coastal defence geometries. The analysis focused on quantifying overtopping volumes and hydrodynamic forces under non-breaking and breaking wave conditions. This study aimed to identify the design trade-offs and propose an optimal geometrical configuration for resilient coastal protection.

1.2 Aims and Objectives

This project aims to investigate the performance of various coastal defence configurations using Smoothed Particle Hydrodynamics (SPH) in DualSPHysics, with a particular focus on wave loading and overtopping volume.

Objectives:

- To establish a stable SPH methodology for simulating wave-structure interactions by validating it against published methods, targeting mean errors below 5%.
- To implement numerical wave flume simulations for various coastal defence geometries such as a vertical seawall, a revetment, and hybrid designs.
- To evaluate structural performance in terms of horizontal loading and overtopping discharge under different wave conditions, by comparing with EurOtop limitations.
- To identify the advantages and limitations of each design, and explore an optimal coastal defence strategy.

2 Literature Review

2.1 Wave Dynamics and Coastal Interaction

Waves are disturbances that propagate through a medium, transferring energy from one point to another in an oscillatory manner. Figure 1: Fundamental Parameters of Wave (Administration 2008) Figure 1 shows the fundamental parameters used in developing analytical wave theories and understanding wave transformation processes. Wave height, H is defined as the vertical distance between wave crest and trough; wavelength, L is the horizontal distance between two consecutive wave crests. The wave frequency, f represents the number of waves passing a point per unit time and is the inverse of wave period, T . The wave celerity, C denotes the speed at which a wave crest propagates and can be related with $C = \frac{L}{T}$. The still water level (SWL) represents the mean undisturbed water surface and serves as the reference datum for coastal structure design.

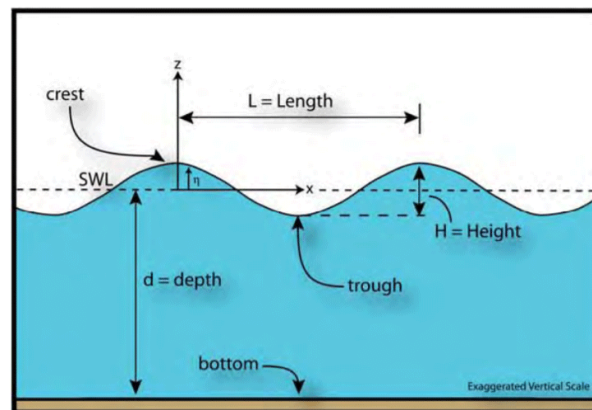


Figure 1: Fundamental Parameters of Wave (Administration 2008)

Linear wave theory, also known as Airy wave theory, is the fundamental analytical approach for describing small-amplitude surface gravity wave behaviour. It assumes an inviscid, incompressible, and irrotational fluid, with small wave steepness and gravitational force as the only external force acting on the system (Holthuijsen 2010). Under these assumptions, the governing equations reduce to Laplace's equation for velocity potential with appropriate boundary conditions. This yields the dispersion relation linking wave speed to depth and an expression for wave energy as a function of wave height. Detailed derivations are included in Appendix A: Linear Wave Theory.

As waves propagate into shallow water, their amplitude and direction are modified by shoaling, refraction, diffraction, and reflection. Shoaling refers to the increase in wave

height as the waves propagate into shallow water and reduction in group velocity to conserve energy flux. Refraction is the bending of wave crests towards shallower water, causing waves to align more perpendicular to the shoreline. Diffraction involves the lateral spreading of wave energy into sheltered areas, such as behind breakwaters or seawalls. Reflection, on the other hand, occurs when incident wave energy is partially or fully reflected seaward due to presence of steep coastal structures such as vertical walls. This reflected energy can interfere with incoming waves, producing standing wave patterns and may amplify local wave heights in front of coastal defences. Both diffraction and reflection arise from the interaction of waves with built or natural structures and are therefore categorised as wave-structure interactions. These effects can induce substantial erosion on coastal defences and can contribute to structural damage if not adequately considered during the design process. These processes can be accommodated by the linear wave theory, provided that the wave steepness remains low and water is not extremely shallow.

In very shallow water, waves may break when their steepness exceeds stability limits transforming into breaking waves. The distinction between non-breaking and breaking waves is essential to coastal defence design.

- Non-breaking waves: Typically associated with gentle slopes or smaller wave heights. They produce cyclic, oscillatory loading on structures.
- Breaking waves: Occur when wave height becomes large relative to depth where linear wave theory no longer applies. Depending on the foreshore slope these may be categorised as follows:
 - Spilling breakers are found on gently sloping beaches, with the wave crest spilling forward to create foam and turbulent white water.
 - Plunging breakers are found on moderately sloped beaches, with the wave crest curling over and plunging forward, dissipating energy in a concentrated area.
 - Collapsing breakers occur on steep slopes where the wave surges up the structure without fully breaking and leads to violent run-up and high overtopping.

2.2 Coastal Defence Systems

2.2.1 Hard Engineering Surfaces

Hard engineering represents the conventional approach to coastal protection, using artificial, static structures that act as an impenetrable barrier between the land and the sea. It is designed to provide immediate resistance against wave attack and flooding. These defences are typically constructed from concrete, rock, steel, or timber and are designed to resist horizontal wave loading while minimising overtopping. Hard defences in the UK are commonly categorised into shore parallel structures, shore normal structures, offshore structures and cliff stabilisation systems. Their effectiveness is typically assessed using two main criteria namely hydrodynamic forces acting on the structure and overtopping discharge across the crest (Van der Meer 2018). However, despite their wide application, hard engineering approaches are increasingly scrutinised due to their environmental impact, high construction and maintenance costs, and inflexibility in the face of rising sea levels and dynamic coastal processes.

Seawalls are the most common among all the shore parallel defences. They present a physical barrier between land and sea, protecting the coastline from erosion and flooding. However, their rigid nature prevents the coastline from responding to environmental changes and can result in significant unintended consequences over time. Most seawalls are vertical or near vertical, which means they reflect rather than dissipate wave energy, resulting in high horizontal forces and elevated overtopping rates. In the UK, seawalls are often constructed on sandy beaches with foreshore slopes typically around 1:20 slope, though slopes can range between 1:15 to 1:100 (Wallingford 2005). This foreshore slope plays a critical role in wave transformation prior to impact, influencing both structural loading and overtopping rates. The reflective nature further exacerbates turbulence and toe scour, leading to structural destabilisation and increased maintenance requirements. Moreover, seawalls contribute to beach degradation by interrupting natural sediment transport processes, leading to beach narrowing. A well-documented example is the former 11km Blackpool seawall frontage, which experienced these issues prior to its redevelopment (French 2001).

Revetments are another common shore-parallel defence, often used alongside or as an alternative to seawalls. Unlike vertical walls, revetments are constructed as sloped structures from durable materials such as concrete armour units, rock, or blockwork. The

sloping geometry is designed to dissipate wave energy, substantially reducing the magnitude of horizontal forces acting on the structure compared to vertical walls. The gradient of the slope of the rubble mound is usually set at 1:1.5 to 1:3 (Goda 2010). A typical design consists of an armour layer, filter layers, and toe protection. The armour layer forms a geometric pattern, filter layers ensure drainage and protection of the underlying soil, and finally toe protection offers stability against undermining at the base of revetment (Staff 2018) and maintains a degree of adaptability to changing beach morphology. While their performance remains sensitive to crest elevation, revetments are more adaptive to morphological change and are often regarded as more sustainable than traditional seawalls.

Hybrid structures that combine vertical walls with seaward sloping revetments aim to capitalise on the strengths of both systems. The revetment slope induces wave breaking and reduces impact loads, while the vertical wall or parapet provides additional crest freeboard to limit overtopping. This configuration is applied in countries with high wave exposure with limited hinterland space such as Japan, the Netherlands, as well as the United Kingdom (Raby 2015),(Van der Meer 2018), (Ishimoto 2014). However, these hybrid structures require greater construction complexity than standalone seawalls and revetments.

Offshore breakwaters are another common shore-parallel structure placed seaward of the beach. It is designed to reduce incoming wave energy and promote sediment deposition in the sheltered area, inspired by natural formations such as offshore islands and reefs, these natural formations function by dissipating wave energy before it reaches the shoreline (Brampton and Smallman 1985). They are generally constructed out of concrete or stone and can either be emergent or submerged below the water surface. Their performance depends on various factors such as crest height, spacing, and distance from shore. While effective in moderate energy environments, they require careful design to avoid flanking erosion and disruption of longshore sediment transport. The Elmer breakwater system in West Sussex has led to significant beach accretion and improved shoreline stability (Agency 2018).

Shore normal structures such as groynes and jetties are constructed perpendicular to the coastline to manage sediment transport. They interfere with longshore drift by trapping sediment, causing accretion on the updrift side and erosion on the downdrift side. While

groynes can extend beach width and provide effective local protection, they often result in sediment imbalance along the coast. Groynes are often made from timber, rock, or concrete, and are frequently used in sediment-deficient coastal systems like in Bournemouth (Council 2022).

Table 1 shows a visual summary of selected hard engineering defences discussed above, including images of typical seawalls, revetments, hybrid structures, offshore breakwaters, and groynes. These structures are integrated to coastal protection systems around the globe due to their ability to resist high-energy wave action and provide immediate defence for vulnerable hinterlands.

Table 1: Hard Engineering Structures

Seawall (Shark Research & Conservation Program n.d)	
Revetments (Wiki n.d.)	

Hybrid Structures (contributors 2025)	
Offshore Breakwater (Pan n.d.)	
Groynes (contributors n.d.)	

Despite their effectiveness, a critical limitation of all hard engineering defences is their susceptibility to wave overtopping, where wave crest exceeds the crest elevation of the structure and allow water to flow landward. Overtopping can lead to localised flooding, structural deterioration and damage to assets behind the structures. It is not only a critical failure mode but also serves as a primary design criterion in assessing the hydraulic performance and effectiveness of coastal defences. The risk of overtopping

depends on factors such as wave height, period, structural geometry and freeboard. Empirical design tools such as the EurOtop Manual provide widely accepted methods for predicting overtopping discharges and defining safety thresholds for various structure types (Van der Meer 2018). Consequently, overtopping performance has become a central concern in coastal defence design and is often employed to evaluate the long-term resilience and reliability of coastal structures.

In response, there has been a growing trend of replacing hard defences with soft engineering or hybrid approaches (French 2001). The UK's Flood and Coastal Erosion Risk Management Strategy promotes the integration of nature-based solutions and hybrid approaches as part of a long-term, adaptive framework. This shift reflects a broader recognition of the need for coastal strategies that are sustainable, resilient and ecologically sensitive (Agency 2020).

2.2.2 Soft Engineering Surfaces

Over the last few decades, there has been a gradual transition from hard to soft coastal defence techniques. As discussed previously, hard defences aim to protect the hinterland but do not contribute to maintaining sediment supply or a natural beach profile. In contrast, soft engineering approaches such as beach nourishment, dune building, saltmarsh restoration and managed realignment enable the coastline to respond dynamically to environmental conditions. These methods enhance the natural coastal processes that dissipate wave and tidal energy and are often more cost-effective and ecologically sensitive than traditional hard structures. Appropriate application of soft engineering solutions should improve the long-term sustainability, adaptability and resilience of coastal defences (Richardson 1996). Each approach has a specific role, and modern coastal management strategies rely on hybrid strategies that combine both hard and soft approaches.

Beach nourishment is one of the most efficient techniques for mitigating coastal erosion and dissipating wave energy. It involves depositing large amounts of replacement sediment to a threatened or eroded shoreline. It is often regarded as the preferred alternative to hard engineering due to its lower environmental impact and the ability to maintain the natural coastline. A wide, gently sloped beach is more effective at dissipating wave energy than steep, narrow and reflective beaches (French 2001). Despite the benefits, beach nourishment projects are not a permanent solution. Its lifetime is

constrained by sediment transport processes and the frequency and intensity of extreme sea states, and the resilience of the system (Martell 2020). It was believed that in some cases the performance of a beach nourishment project is enhanced by the addition of hard structures (Council 1995).

Sand dunes are unique landforms that are formed by the wind rather than the waves, representing a storage of sand in the supratidal zone. They represent a natural barrier between the land and sea, functioning like a sea wall. The dune face and embryo dunes are dynamic, capable of supplying sediment to the beach during erosive events and storing it during calmer conditions. Dunes can be enhanced by planting vegetation such as Marram grass, which traps the wind-blown sand and increases dune volume size in the order of 1 m a^{-1} (French 2001). However, dunes are fragile systems due to the uncompacted sediment and shallow root structure, vulnerable to damage from overuse and trampling. Figure 2 shows a typical cross-section of a coastal dune system, from embryo dune to fixed inland vegetation. Thus, consistent monitoring and management are essential to maintain their integrity. In Netherlands, dunes serve as the primary defence against coastal flooding, highlighting the effectiveness of dunes as a natural coastal protection system (Magazine 2023).

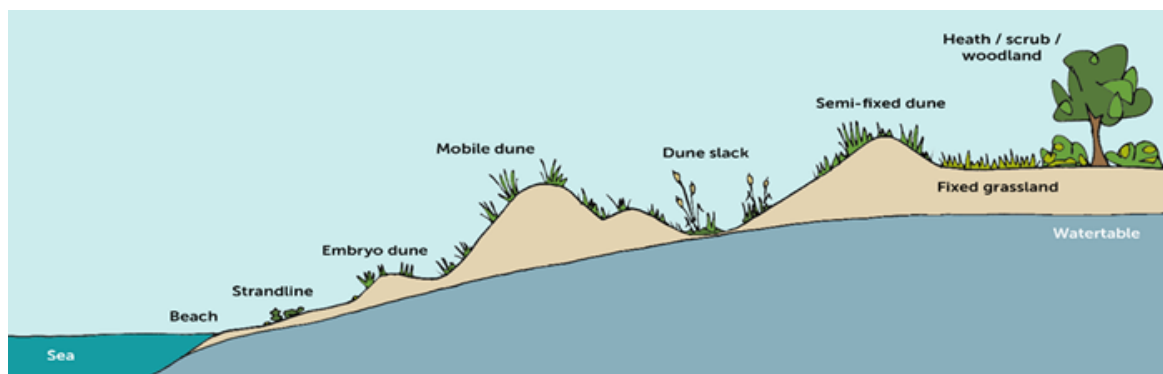


Figure 2: Typical Dune Profile (HaleyOwAndersen 2022)

Saltmarshes are intertidal ecosystems found in low energy temperate coastlines. Composed of salt-tolerant vegetation, they provide an effective natural buffer between land and sea by dissipating wave energy and stabilising sediment (Koch 2009). In addition to their protective role, saltmarshes pose significant ecological value such as supporting biodiversity, improving water quality, and sequestering carbon (Billah 2022). Additionally, when used in conjunction with seawalls, it was shown that it can effectively reduce both construction and maintenance costs, particularly when vegetated marsh

widths exceed 80 metres (Vuik 2016). However, saltmarshes are increasingly threatened by rising sea levels and coastal squeeze, where landward migration is prevented by hard infrastructure. Conservation and restoration are therefore essential to ensure their function in sustainable coastal defence strategies.

Lastly, managed realignment is considered an effective soft engineering approach that involves the deliberate repositioning of river, estuarine, or coastal defences (DEFRA 2002). This can include setting back defences to higher ground, constructing a new line of defence inland or shortening the overall length of coastline requiring active maintenance. The primary aim is to reduce long-term maintenance costs while enhancing the efficiency and sustainability of coastal protection. Managed realignment can also deliver wider environmental benefits such as habitat creation, biodiversity enhancement and improved flood storage capacity.

2.3 Design Standards and Guidance

The design of coastal defence systems is governed by a set of standards and guidelines to ensure structural integrity, economic feasibility, and long-term sustainability. These standards usually address aspects such as hydraulic loading, material selection, geotechnical stability, and resilience under future climate conditions. Organisations such as the U.S. Army Corps of Engineers (USACE) and CIRIA provide widely adopted technical frameworks for the design and performance assessment of coastal infrastructure. In the UK, documents such as the Flood and Coastal Erosion Risk Management (FCERM) Strategy set out long-term objectives for adaptation and coastal planning. Moreover, the CIRIA C683 Coastal and Marine Design Manual offers detailed technical design principles and engineering methodologies for various coastal defences.

In addition, Shoreline Management Plans are implemented to manage flood and coastal erosion risk around the coast (Agency 2024). Government agencies such as the Environment Agency and DEFRA have also introduced multiple programmes to strengthen resilience to flooding and coastal change (DEFRA 2020).

The EurOtop Manual on Wave Overtopping (Van der Meer 2018) is a key reference for the hydraulic performance of hard coastal structures. It provides comprehensive guidance for predicting wave overtopping discharges across a wide range of structures

such as seawalls, breakwaters, and revetments. The manual combined experimental results and numerical models to deliver empirical formulas and design thresholds. Its guidance is widely adopted and highlights the overtopping safety limits employed in various design phases.

Collectively, these standards provide a coordinated approach for designing and implementing coastal defences that are robust, pragmatic, adaptive, and aligned with sustainable development goals.

2.4 Numerical and Experimental Modelling Approaches

The design of effective coastal defences requires accurate prediction of wave run-up, overtopping and impact loading, and is typically assessed through a combination of physical experiments and numerical modelling.

Experimental approaches typically employ wave flumes to create controlled conditions where wave-structure interactions can be systematically observed, measured and analysed. Moller et al. found that the marsh vegetation can attenuate wave energy by up to 60% using a 300m long wave flume (Iris Möller 2014). Moreover, Lorenzoni et al. assessed the fundamental features of various breakwater configurations using a 2D movable-bed flume and found that emerged breakwaters provide superior protection compared to the submerged breakwaters. However, they are also more sensitive to dimensional and positional variations (Carlo Lorenzoni 2016). Le Hai Trung et al. conducted a hydraulic model test in a 45 m wave flume to assess effectiveness of bullnoses on seawalls under irregular waves. It was demonstrated that bullnoses reduced overtopping discharge by 85% and limited splash run-up on vertical seawalls (Le Hai Trung 2020). Neves and Didier further investigated overtopping at a smooth impermeable dike with promenade using a 16 m flume and highlighted the challenges of predicting performance under plunging wave conditions. Together, these experimental studies provide essential benchmarks for validating numerical tools.

While physical tests are invaluable, they are often expensive and time-consuming with limitations to certain structures. Computational methods are increasingly adopted for their ability to simulate highly nonlinear free-surface flows and complex wave-structure interactions. Numerical approaches can be further categorised into:

- Mesh-based CFD: Implemented in solvers such as OpenFOAM, Fluent where it solves the Reynolds-averaged Navier-Stokes (RANS) equation using Volume-of-Fluid (VOF) techniques to capture free surfaces.
- Smoothed Particle Hydrodynamics (SPH): SPH is a meshless CFD method, where it discretises the fluid into particles governed by the Navier-Stokes equations. Its Lagrangian formulation makes it naturally suited to highly non-linear phenomena such as wave run-up, wave breaking and overtopping.

SPH has been employed extensively in the coastal engineering industry. Altomare et al. used DualSPHysics to model wave impact on quay walls and storm return walls, showing good agreement with semi-empirical formulae and flume data (Altomare 2015). Benedict et al. contributed to the development of GPU-accelerated SPH algorithms to enhance computational performance (Benedict et al. 2015). Crespo et al. applied DualSPHysics to simulate wave run-up over coastal structures such as rubble mound breakwater with grooved cubes, validating results against laboratory data (Crespo, Altomare et al. 2013). Ni and Feng extended SPH overtopping and hydrodynamic forces studies to various seawall geometries with mean deviations of only 6.8% from experimental studies (Ni 2013). Moreover, Dang et al. demonstrated with DualSPHysics that stepped-face seawalls significantly reduced overtopping compared to vertical and inclined faces, while maintaining acceptable hydrodynamic force levels (Dang 2021). Barreiro et al. further validated the approach against analytical and experimental benchmarks, establishing confidence in SPH's predictive accuracy (Barreiro 2013).

Nevertheless, CFD grid-based methods remains widely used, particularly for turbulence-resolving simulations. Huang et al. reviewed various wave structure interaction modelling methods using OpenFOAM (HUANG, et al. 2022). Barendse et al. employed OpenFOAM to determine optimised revetment geometries that minimise hydrodynamic loading and improve structural efficiency (Luuk Barendse , et al. 2022).

The SPH method offers a flexible, meshless approach capable of modelling highly nonlinear wave dynamics and free-surface interactions, well suited for coastal engineering problems. Unlike grid-based CFD methods, SPH is a meshless approach, making it highly effective for problems with large deformations, moving boundaries and breaking waves. Accordingly, this dissertation employed SPH, implemented via the open-

source DualSPHysics as the primary numerical framework to simulate wave-structure interactions and to evaluate the optimal design of coastal defence systems.

2.5 Concluding Remarks

This literature review highlighted the wide range of coastal defence strategies, from traditional hard structures like seawalls, revetments and breakwaters to nature-based soft engineering approaches including beach nourishment, dune restoration and saltmarsh enhancement. While soft and hybrid solutions are increasingly promoted for long-term resilience and sustainability, hard engineering surfaces are still widely implemented in high-risk, developed coastal zones. Horizontal hydrodynamic forces and wave overtopping discharge are the two key performance measures in the design of hard defences. Excessive loading can lead to structural instability, while overtopping presents both a safety hazard to the hinterland and failure for coastal infrastructure.

Numerical approaches are generally adopted to evaluate these performance measures. While experimental flume studies provide valuable insights, they are often resource-intensive and limited to specific geometries. Numerical modelling offers a more flexible and efficient means of exploring a broader range of wave-structure interactions. Among all the numerical approaches, the SPH method has been shown to be effective for modelling highly non-linear surface flows and is capable of handling large free-surface deformations without the need for mesh generation.

This study focused on four hard engineering structures: a vertical seawall, a sloped impermeable revetment and two hybrid configurations combining a revetment and seawall of varying height acting as a parapet. While hybrid structures are increasingly implemented in practice, their sensitivity to crest elevation and parapet geometry remains insufficiently studied under systematically controlled conditions. This study employed the open-source DualSPHysics code to simulate the wave-structure interactions on the proposed geometries in a two-dimensional wave flume. The analysis focused on quantifying both horizontal forces and overtopping performance across the four different structures to identify an optimal configuration that balances structural loading with overtopping resilience.

3 Methodology

3.1 Introduction

Smoothed Particle Hydrodynamics (SPH) was originally developed in the 1970s for astrophysical and cosmological applications (Barreiro 2013). In recent years, it has gained significant traction within the fluid dynamics and coastal engineering fields, due to its ability to model complex free-surface flows with high geometric flexibility.

SPH is a fully Lagrangian, meshless computational method in which the fluid is discretised into a set of particles. Each particle carries physical properties such as density, mass, velocity, and pressure, and evolves over time according to the Lagrangian form of the Navier-Stokes equations. Unlike conventional mesh-based computational fluid dynamics (CFD) techniques, such as finite-volume, finite-difference or finite-element methods, SPH naturally handles large deformations, free-surface motion, and complex fluid-structure interactions without a predefined computational mesh. This renders it particularly well-suited for problems involving moving boundaries, wave breaking, and fluid-structure interaction.

This section outlines the formulation of the SPH method and specific implementation considerations using the DualSPHysics software relevant to wave-structure interaction problems.

3.2 Smoothed Particle Hydrodynamics

3.2.1 Interpolation Integral

The fluid domain in the SPH method is discretised into a set of particles. The foundation of SPH method lies in integral interpolation, where an arbitrary scalar field $A(r)$ is approximated as weighted average over neighbouring particles using a smoothing kernel. This is mathematically expressed as:

$$A(r) = \int_{\Omega} A(r') W(r - r', h) dr' \quad 3.1$$

where r is the position, W is the smoothing kernel, h is the smoothing length that defines the extent of the kernel's compact support domain Ω . The kernel acts as a compactly supported weighting function, assigning higher influence on particles closer to r .

The integral is discretised over the set of neighbouring particles b , yielding the SPH approximation:

$$A(r_a) = \sum_b A(r_a) W(r_a - r_b, h) \frac{m_b}{\rho_b} \quad 3.2$$

mass and density particles are denoted by m_b and ρ_b respectively, and W is the kernel evaluated at the distance between particle a and each neighbour b . This discretisation allows field quantities to be approximated in a mesh-free manner using only particle data and kernel functions.

3.2.2 Kernel Function

The choice of the weighting function depicts the numerical accuracy and stability of the SPH model. These functions must satisfy conditions such as positivity, compact support and normalisation to ensure convergence and consistency. Moreover, the kernel function W must decrease monotonically with increasing particle distance and approximate a Dirac delta function as the smoothing length, $h \rightarrow 0$ (Monaghan 1992)

DualSPHysics framework supports two kernel options namely the cubic spline kernel, which is widely used due to its simplicity and computational efficiency, and the Wendland kernel, which is utilised in this study (Crespo 2015). The kernel adopted in this work is the quintic kernel presented by Wendland (Wendland 1995) as:

$$W(q, h) = \alpha_D \left(1 - \frac{q}{2}\right)^4 (2q + 1) \quad 0 \leq q \leq 2 \quad 3.3$$

where $q = r/h$ and $\alpha_D = 21/(16\pi h^3)$ in 3D. The Wendland kernel is favoured for its higher-order continuity and improved stability in pressure and density estimation. It is well-suited for simulations involving strong free-surface motion and high gradients, as commonly encountered in wave-structure interaction (Majtan 2022).

3.2.3 Governing Equations

In the SPH method, the governing equations of fluid motion are formulated in a Lagrangian framework using particle-based interpolation schemes. The key equations are the continuity equation (mass), momentum equation, and the equation of state. These are discretised using SPH approximation and solved explicitly over time.

Continuity Equation

The continuity equation governs the conservation of mass. The mass of each particle is constant, and the rate of change of fluid density is computed from the relative velocities of neighbouring particles in SPH as follows:

$$\frac{d\rho_a}{dt} = \rho_a \sum_b v_{ab} \cdot \nabla_a W_{ab} \frac{m_b}{\rho_b} \quad 3.4$$

where $v_{ab} = v_a - v_b$ and the m_b is the constant mass of each particle b and $\nabla_a W_{ab}$ is the gradient of kernel function evaluated at the position of particle a .

Momentum Equation

The momentum equation describes the conservation of linear momentum and involves pressure and gravitational forces, as well as the artificial viscosity to enhance stability. The momentum equation proposed by Monaghan (Monaghan 1992) is rewritten into SPH notation as

$$\frac{dv_a}{dt} = - \sum_b m_b \left(\frac{P_b}{\rho_b^2} + \frac{P_a}{\rho_a^2} + \Pi_{ab} \right) \nabla_a W_{ab} + g \quad 3.5$$

where P is pressure and g is vector of gravitational acceleration. The weighting function W_{ab} depends on distance between particles a and b . The term Π_{ab} is the viscous term depending on the artificial viscosity of flow as:

$$\Pi_{ab} = \begin{cases} -\frac{\alpha \overline{c_{ab}} \mu_{ab}}{\overline{\rho_{ab}}} & v_{ab} \cdot r_{ab} < 0 \\ 0 & v_{ab} \cdot r_{ab} > 0 \end{cases} \quad 3.6$$

where $v_{ab} = v_a - v_b$ and $r_{ab} = r_a - r_b$, which are the particle velocity and positions, h is the smoothing length, $\mu_{ab} = h v_{ab} \cdot r_{ab} / (r_{ab}^2 + \eta^2)$ with $\eta^2 = 0.01 h^2$; $\overline{c_{ab}} = 0.5(c_a + c_b)$ and $\overline{\rho_{ab}} = 0.5(\rho_a + \rho_b)$ are the mean speed of sound and density respectively. The $\alpha = 0.01$ is the artificial viscosity coefficient utilised to enhance the numerical stability.

Equation of State

DualSPHysics adopts a weakly compressible SPH formulation (Majtan 2022) where the pressure is related to density through an equation of state. The Tait equation is used to model fluid compressibility:

$$P = \frac{c_o^2 \rho_o}{\gamma} \left[\left(\frac{\rho}{\rho_o} \right)^\gamma - 1 \right] \quad 3.7$$

where $c_o = c(\rho_o) = \sqrt{(\partial P / \partial \rho)|_{\rho_o}}$ is the speed of sound at the reference density, $\rho_o = 1000 \text{ kg m}^{-3}$, and the polytropic constant, $\gamma = 7$. The value of c_o is chosen to ensure that density variations remain below 1%, preserving the assumption of weak compressibility while keeping a stable numerically stable (Monaghan 1992).

3.2.4 Time-Stepping Scheme

In SPH, the physical properties of each particle evolve over time due to particle interactions. The time integration algorithms must be at least second order to accurately integrate the governing equations. DualSPHysics offers two integration schemes namely Verlet and Symplectic.

Symplectic integration scheme is employed in this study due to its increased robustness for long-term simulation. The scheme features a predictor-corrector structure and is time-reversible in the absence of dissipative effects such as viscosity and friction (Leimkuhler 1996). The Symplectic algorithm exhibits greater numerical stability by Crespo (Crespo 2015). The time-step size is automatically adjusted at each iteration according to the Courant-Friedrichs-Lewy (CFL) condition, particle interaction speeds, and viscous constraints. This adaptive control ensures stable and accurate time integration throughout the simulation.

3.2.5 Boundary Conditions

Boundary conditions are enforced through particle-based methods rather than analytical expressions in SPH. DualSPHysics defines solid boundaries by assigning material markers such as `<mkbound>` to specific regions within the geometry, enabling simulation of static walls, moving paddles, and floating structures. The boundary particles interact with the fluid particles via the same governing equations.

In this study, boundary conditions were implemented through the definition of static walls and a moving piston-type wave paddle using the `<drawbox>` and `<drawprism>` commands in the XML input file. The wave paddle motion was prescribed via the `<motion>` command and was defined as a second-order piston-type wave generator. Active Wave Absorption System (AWAS) was employed to minimise reflective wave interference at the paddle boundary.

3.2.6 DualSPHysics Implementation

The open-source computational tool DualSPHysics (<https://dual.sphysics.org/>) was employed in this study to model free-surface flows using the SPH method on Graphics Processing Units (GPU). The code originates from SPHysics, a collaborative project developed by researchers at the Johns Hopkins University (USA), the University of Vigo (Spain) and the University of Manchester (UK). DualSPHysics is written in C++ with parallelisation achieved through NVIDIA CUDA, enabling efficient large-scale particle simulations. The code has been widely applied to a range of fluid-structure interaction problems, including dam-break flows, wave impact scenarios, and landslides. It has also demonstrated utility in industrial settings, where it was used to simulate free-surface flows on the Pelton turbine (Marongiu 2010).

SPH-based simulation in DualSPHysics generally follows a three-stage workflow: pre-processing, processing and post-processing.

Pre-processing

During the pre-processing stage, DualSPHysics defines the computational domain, particle characteristics, geometry, and SPH physical parameters via a structured XML input file. This configuration file is composed of two main blocks namely, <casedef> and <execution>, which describe the initial conditions and numerical simulation controls, respectively.

The <casedef> section governs the specification of SPH constants, including gravitational acceleration, reference density, speed of sound, smoothing length, and Courant-Fredrichs-Lewy (CFL) number. These parameters were selected based on established SPH practice for free-surface flows. The simulation geometry, such as fluid regions, wave paddles, and structural boundaries is defined with unique material markers within the <casedef> section as well. In addition, <casedef> may include optional elements such as <initials> for initial particle states, <floatings> for floating object description, and <motion> for prescribing boundary movement. This section is constructed by the user and processed by the GenCase utility which generates the corresponding particle distribution and configuration files.

The <execution> block defines numerical control parameters required for executing the simulation. It specifies the wave generation settings, numerical schemes, viscosity and kernel models, and solver execution parameters such as time-stepping and output frequency. This block is used exclusively by the DualSPHysics solver and is partly modified during the pre-processing stage.

Processing

The simulation is executed using the DualSPHysics solver via a command-line interface. DualSPHysics supports both CPU and GPU computation, with GPU acceleration enabled in this study to facilitate high-resolution, large-scale simulations. Simulation outputs, including velocity, pressure, and free-surface elevation, are recorded at regular intervals specified within the XML configuration file. These data are subsequently used for post-processing analyses, including force evaluation and surface elevation tracking.

Post-processing

Post-processing was carried out using tools provided within the DualSPHysics framework. Visualisation was performed using ParaView (v6.0.0) with the .vtk files

generated from the solver output. These files contain time-resolved data on both fluid and boundary particle states.

Quantitative analyses were conducted using ComputeForces to evaluate hydrodynamic forces on the structure, MeasureTool to extract pressure and velocity data at predefined probe locations, and FlowTool to calculate overtopping discharge across a predefined domain. These quantitative tools were executed via a user-defined batch script, which also handles simulation and post-processing stages. The script was edited to configure each tool, including marker numbers, probe locations, and output settings for force, pressure, and overtopping analyses.

3.3 Benchmark Case

A benchmark case was reproduced from the study (Dang 2021) which investigated wave loading on various coastal defence geometries using SPH in order to validate the suitability and accuracy of the applied methodology in this study. A convergence study was conducted as well to ensure that the selected particle spacing and time step provided resolution-independent results, thus confirming the numerical stability and accuracy of the simulation setup. The reference study employed a 2D wave flume measuring 7.0 m in length and 1.5 m in height model, as shown in Figure 3. Regular waves were generated by a piston-type paddle using second-order wave theory, with both non-breaking and breaking wave conditions modelled, as summarised in Table 2.

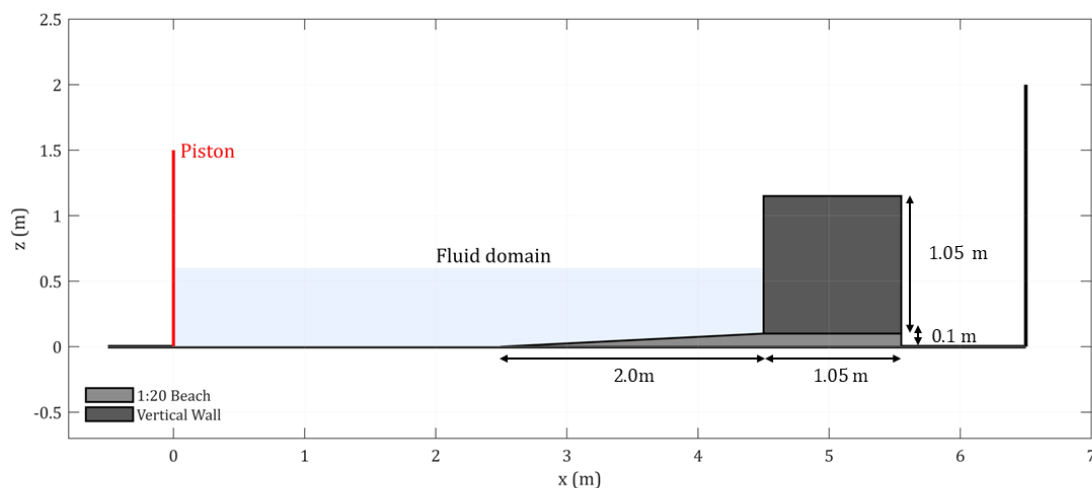


Figure 3: Benchmark Case Domain

The vertical wall (VW) configuration from the reference study was simulated for four wave conditions. The seawall model was placed over a rigid and impermeable bed of

slope 1:20. The horizontal hydrodynamic forces on the vertical wall were computed using the ComputeForces tool in DualSPHysics, with the peak force value extracted for each simulation. The results were compared with the corresponding values reported in the referenced numerical study, and the analytical estimates, providing a quantitative benchmark for the present SPH method.

Table 2: Wave Condition Parameters

Wave Parameters	Non-breaking waves		Breaking Waves	
	G1	G2	T1	T2
$H [m]$	0.10	0.11	0.25	0.27
$T [s]$	2.00	2.20	2.00	2.10
$d [m]$	0.60	0.6	0.60	0.60

3.4 Run Case

Upon reviewing the literature review, four 2D case studies were proposed to investigate the optimal design of hard defences. The numerical experiments were carried out in a wave flume with a fixed height of 1.5 m, which extended 1.0 m landward of each structure. Four coastal defences were studied, namely a vertical wall, a revetment with a slope ratio of 1:1.5 which falls within the standard design range of a revetment (Goda 2010), and two hybrid configurations including revetment and a parapet wall of different heights. For a consistent comparative assessment, the total structure height across all cases was standardised at 0.9 m, with each structure placed on a foreshore slope of 1:20. Figure 4, Figure 5, Figure 6, Figure 7 show the computational domain with exact dimensions of each case. The same four wave conditions listed in Table 2 were employed for all case studies. Wave generation was implemented using a piston-type paddle with second-order wave theory in DualSPHysics. A particle distance of $\Delta p = 0.01 \text{ m}$ was employed in the run cases, resulting in approximately 37,000 particles in the domain.

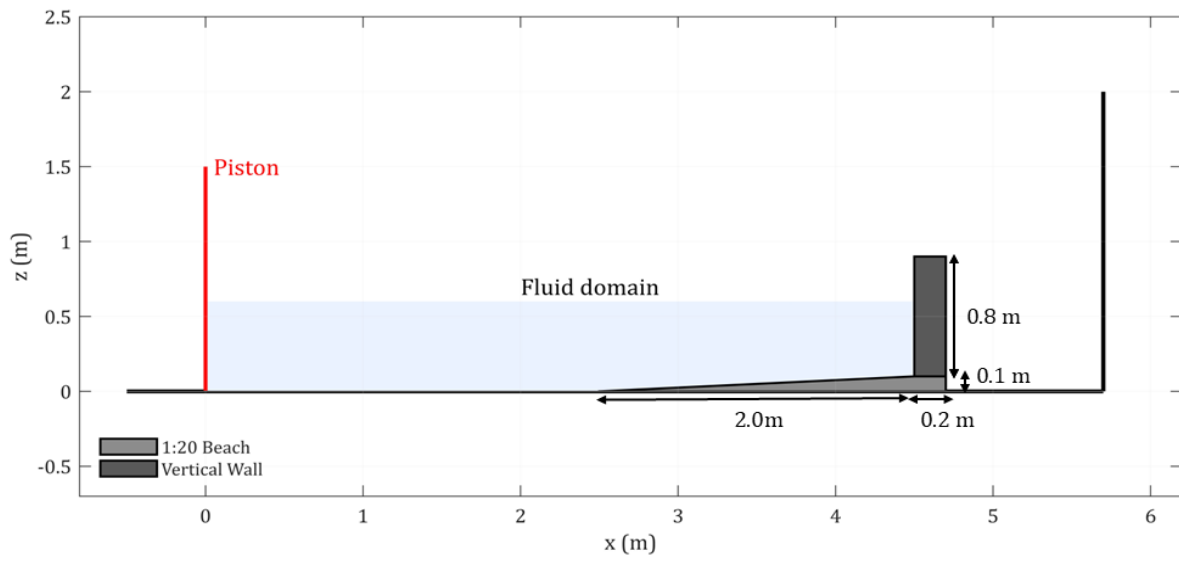


Figure 4: Vertical Seawall Domain

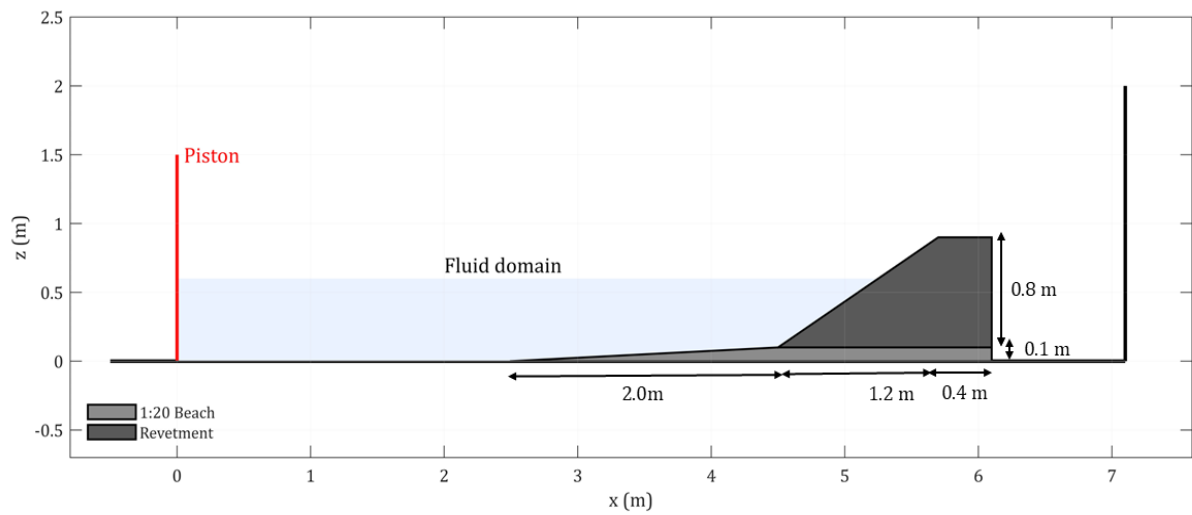


Figure 5: Revetment Domain

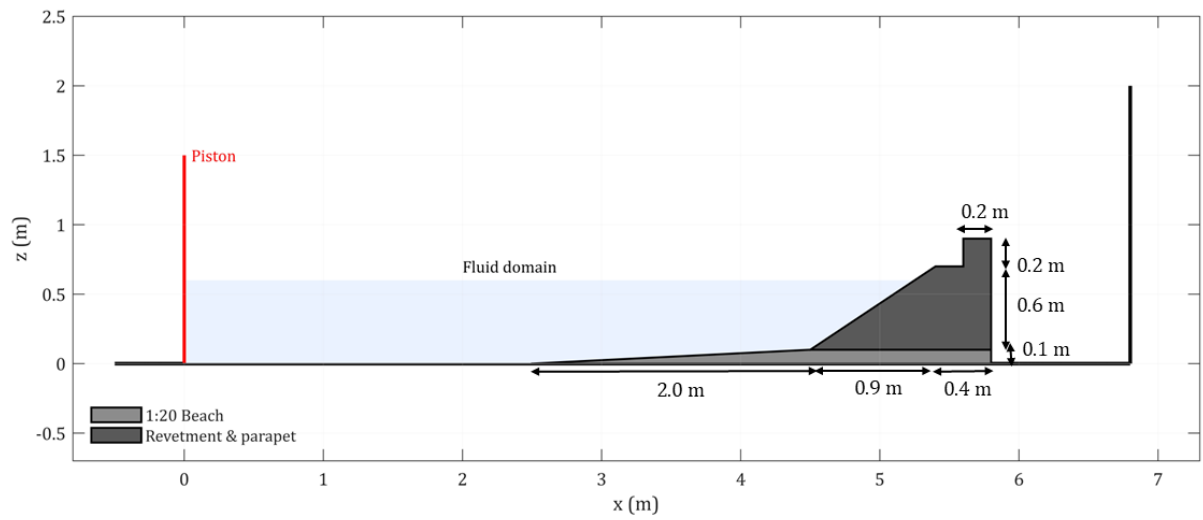


Figure 6: Hybrid A domain (Revetment 0.1 m above SWL)

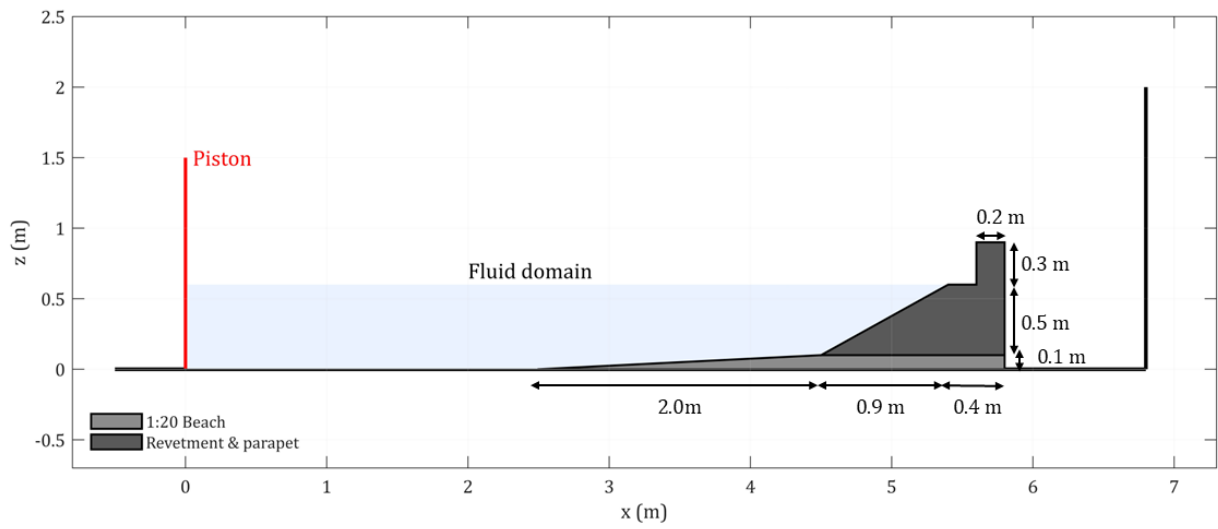


Figure 7: Hybrid B domain (Revetment at SWL)

4 Results and Discussion

4.1 Benchmark Case

This section presents the results for the vertical wall (VW) benchmark case, simulated under four wave conditions summarised in Table 2. Horizontal hydrodynamic forces were extracted from the DualSPHysics simulations and compared with the values reported by Dang (Dang 2021) as well as with analytical estimates using Goda’s formula for vertical walls (Goda 1967).

4.1.1 Convergence Study

A spatial convergence test was performed for the non-breaking wave G1 case by refining the particle spacing Δp from 0.020 m to 0.005 m, while keeping all other numerical settings constant. The maximum horizontal force from the SPH simulations was compared against both the published reference SHP result and the analytical estimate by Goda (Goda 1967). Table 3 shows that the change in force between $\Delta p = 0.01$ m and $\Delta p = 0.005$ m was less than 0.2%, indicating the solution was spatially converged. A particle spacing of $\Delta p = 0.01$ m was therefore adopted for all subsequent simulations with $H/dp \approx 10$, as it provided a good balance between accuracy and computational efficiency.

Table 3: Convergence Study Results

Particle Spacing [m]	Present SPH [N/m]	Reference SPH [N/m]	Analytical [N/m]	% Difference (Present vs Reference)	%Difference (Present vs Analytical)
0.020	1718.71	1640.00	1734.70	4.8%	0.9%
0.010	1702.44			3.8%	1.9%
0.005	1704.60			3.9%	1.7%

The horizontal hydrodynamic forces for are presented in Table 4. Results from the present SPH simulations showed close agreement with both the reference numerical study and the analytical estimates, with mean differences of 1.9% and 1.4%, respectively. Non-breaking waves exhibited differences below 5.5%, while the error for breaking waves remained below 2%. Since analytical estimates were derived from established empirical formulations, they provided a robust baseline for validation, and the present

SPH results demonstrated particularly close agreement with these values. Overall, the benchmark results confirmed the reliability of the numerical setup, which was therefore adopted for the subsequent design simulations.

Table 4: Comparison of Hydrodynamic Forces of Benchmark Case

Wave Conditions	Present SPH [N/m]	Reference SPH [N/m]	Analytical [N/m]	% Difference (Present vs Reference)	%Difference (Present vs Analytical)
G1	1702.44	1640.00	1734.70	3.8%	1.9%
G2	1788.16	1700.00	1817.00	5.2%	1.6%
T1	2510.51	2495.00	2554.30	0.6%	1.7%
T2	2709.73	2760.00	2725.20	-1.8%	0.6%
			Mean Error	1.9%	1.4%

4.2 Run Case

This section presents the hydrodynamic response of the four proposed defence configurations. Each structure was assessed with two main performance measures identified in the literature review: (i) maximum horizontal force and (ii) overtopping volume. Comparisons between cases highlighted the influence of the structure geometry on both loading and overtopping performance.

4.2.1 Horizontal Force

Table 5 presents the extracted maximum horizontal forces of the proposed defence surfaces across the four wave conditions. The revetment consistently reduced the maximum loading relative to the vertical wall with reductions ranging from 7% to 16%. In contrast, the hybrid geometries reduced the force under non-breaking waves but exhibited amplified loads under breaking conditions.

Table 5: Maximum Horizontal Forces of Run Cases

Maximum Horizontal Forces [N/m]				
Wave conditions	Wall	Revetment	Hybrid A	Hybrid B
G1	1702.44	1585.02	1608.61	1757.58
G2	1788.16	1638.33	1652.90	1611.59
T1	2593.93	2556.00	2732.19	2885.80
T2	2680.54	2261.50	3338.15	4541.94

To explore the hybrid structures' performance further, the wave loading forces were decomposed into components of revetment and parapet contributions. Table 6 shows the force decomposition for the wave condition T2 at $t = 14\text{s}$, the total force equalled the sum both components confirming the relation $F_{whole} = F_{parapet} + F_{revetment}$.

Table 6: Hybrid B Force Contributions

Components	Forces
Revetment	2753.41 N/m
Parapet	1788.53 N/m
Whole	4541.94 N/m

Table 7 compares the maximum component forces across the simulation. Hybrid A's parapet experienced 46% lower loading than Hybrid B's, owing to its revetment crest being positioned 0.1 m above the SWL of 0.6 m, which dissipated energy before reaching the parapet. Conversely, Hybrid B's revetment crest remained at the same level with the SWL, leading to direct wave-structure interaction on the parapet.

Table 7: Force Comparison of Both Hybrid Structures.

Wave Conditions	T2	
Components	Hybrid A	Hybrid B
Revetment	2557 N/m	2784 N/m
Parapet	1719 N/m	2753 N/m
Whole	3338 N/m	4541 N/m

Figure 8 illustrates the time history of horizontal forces under T2 for both hybrid cases, showcasing that Hybrid A generally experienced lower total force loading.

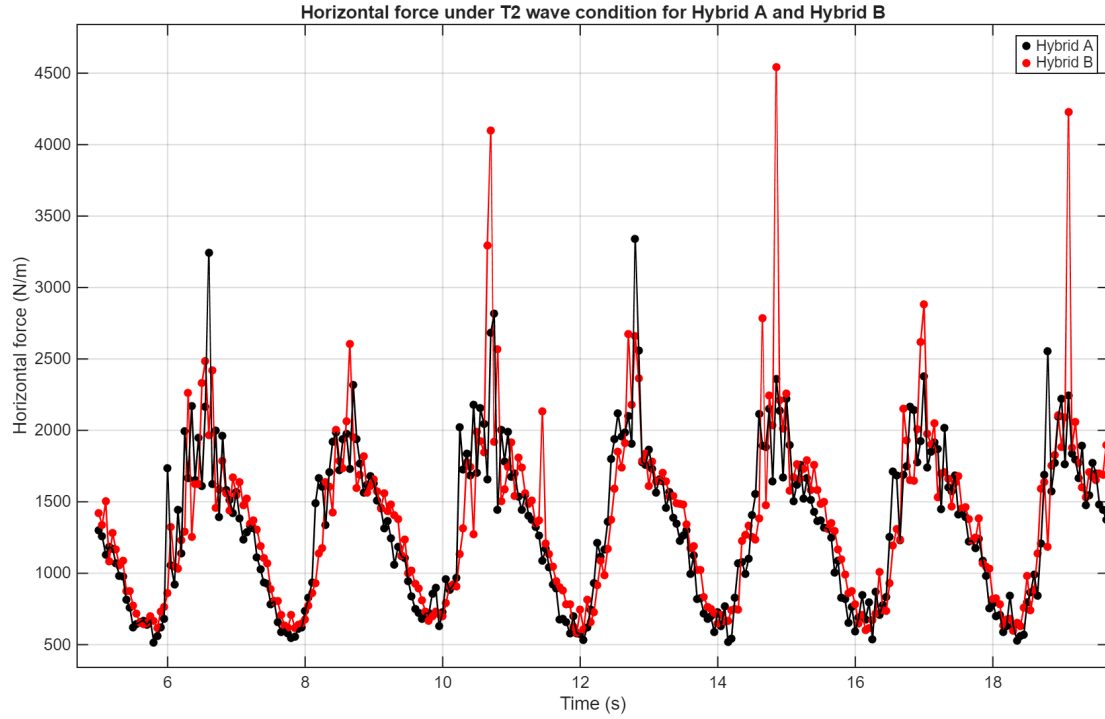


Figure 8: Comparison of Horizontal Force under T2 for Both Hybrid Cases.

4.2.2 Overtopping Volume

The overtopping volume results were computed using a landward collection box positioned 0.05 m behind the structure with a constant 1.0 m depth for all geometries. As shown in Table 8, no overtopping occurred for all non-breaking wave conditions, while overtopping was evident in all breaking cases. The vertical wall case exhibited the lowest overtopping volume at 0.0264 m^3 while the revetment recorded the highest overtopping volume of 0.1684 m^3 at wave condition T2. In contrast, Hybrid A decreased the overtopping volume by 64% to 0.0613 m^3 relative to the revetment, while Hybrid B achieved the greatest reduction in overtopping volume, by 77% to 0.0393 m^3 .

Table 8: Overtopping Volume of Run Cases

Wave Conditions	Overtopping Volume, [m^3]			
	Wall	Revetment	Hybrid A	Hybrid B
G1	0.0000	0.0000	0.0000	0.0000
G2	0.0000	0.0000	0.0000	0.0000
T1	0.0072	0.1012	0.0084	0.0072
T2	0.0264	0.1684	0.0613	0.0393

Moreover, the mean overtopping discharge was calculated using the mean overtopping discharge equation:

$$q = \frac{V_{net}}{TB} \quad 4.1$$

where q is the mean overtopping discharge, V_{net} is the net overtopping volume in m^3 , T is the simulation duration, and B is the structure width. According to the EurOtop Manual, the tolerable overtopping limit for rubble mound breakwaters is approximately 1L/s per m, which corresponds to a threshold of $0.04 m^3$ over the duration of the simulated case studies (Van der Meer 2018). When assessed against this criterion, the revetment case clearly exceeded the safety limit, while Hybrid B approached close to the limitations. In contrast, the vertical wall and Hybrid A remained well below the suggested overtopping discharge limit.

4.2.3 Velocity Flow Fields

The instantaneous velocity contours were extracted at $t = 10, 15$, and 20 s under the T2 wave condition to provide further insight into the hydrodynamic response of the four defence configurations. These instantaneous contours help visualise the various wave-structure interaction and highlight the differences of each configuration.

Vertical Wall

Figure 9 shows the instantaneous velocity flow field of the vertical wall configuration. Across all three instants, the flow remained periodic, with relatively low free-surface velocities and minimal breaking. At $t = 15$ s, reflection from the vertical wall generated a standing-wave pattern. At $t = 20$ s, the velocity remained relatively weak compared to other configurations, with limited overtopping volume visible landward of the structure, consistent with the recorded value. The fluid decelerated rapidly at the wall, indicating high stagnation pressure generated, which explained the higher horizontal forces measured for the geometry.

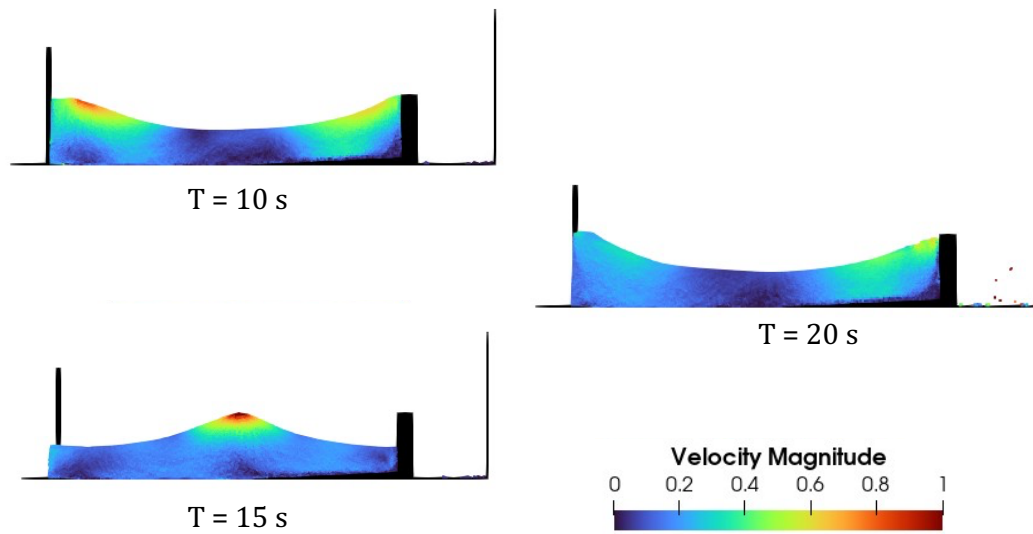


Figure 9: Instantaneous Velocity of Vertical Seawall

Revetment

Figure 10 shows the instantaneous velocity fields for the revetment configuration at $t = 10$, 15 , and 20 s. High velocities were observed along the slope from $t = 10$ s onwards. At $t = 15$ s, the run-up was intensified with significant overtopping over the upper slope and landward of the structure consistent with the large overtopping volume recorded. Unlike the vertical wall, the revetment dissipated energy gradually along its slope, reducing the maximum horizontal forces experienced.

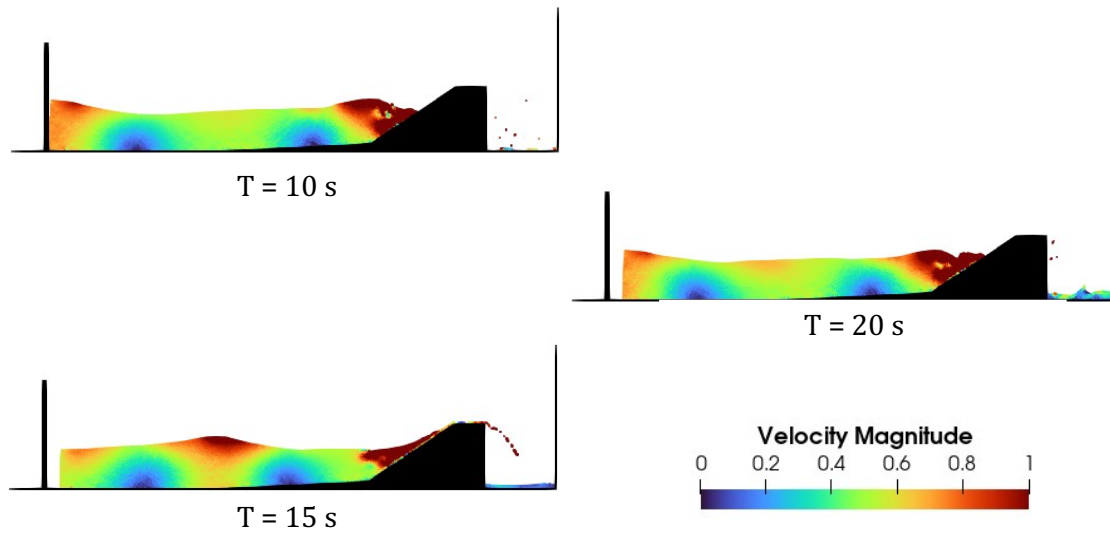


Figure 10: Instantaneous Velocity of Revetment

Hybrid A (Revetment crest above SWL)

Figure 11 shows the instantaneous velocity for Hybrid A at $t = 10$, 15 and 20 s. The overall velocity magnitude was similar to the revetment case with visible run-up along the slope. At $t = 10$ s, the wave run-up was successfully contained by the parapet with flow redirected back down the slope rather than propagating landward. At $t = 15$ s, the overtopping volume was greatly reduced compared to the revetment case. At $t = 20$ s, the landward overtopping volume remained limited, confirming the effectiveness of a parapet. These results aligned with the overtopping measurements, which showed that the Hybrid A structure significantly reduced the overtopping volumes while dissipating energy along the slope to lower parapet loading.

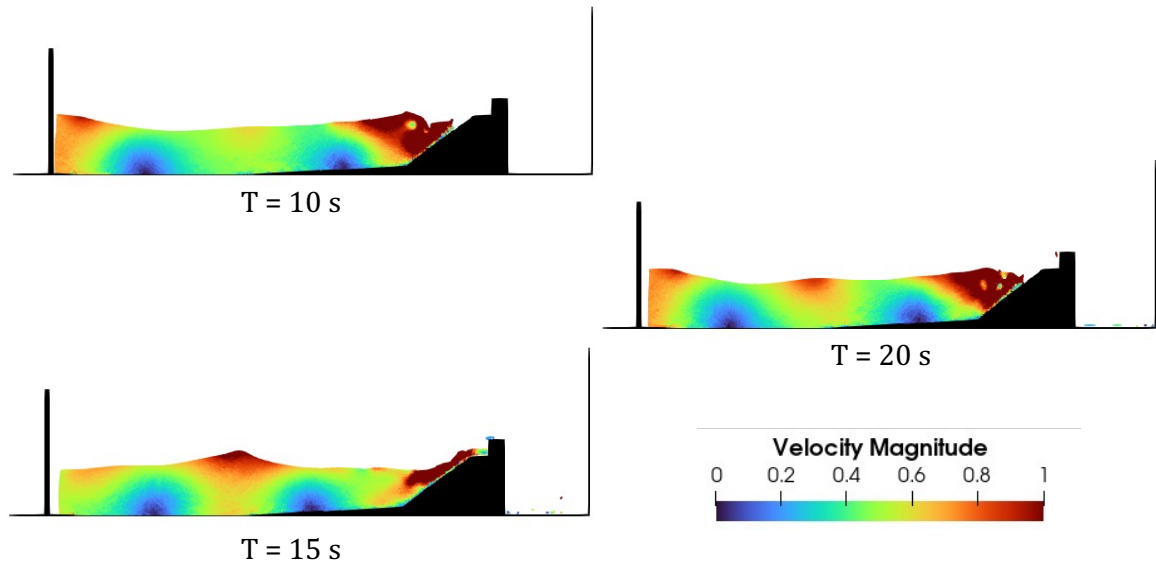


Figure 11: Instantaneous Velocity of Hybrid A (Revetment 0.1 m above SWL)

Hybrid B (Revetment crest at SWL)

Figure 12 shows the instantaneous velocity fields for Hybrid B at $t = 10$, 15 and 20 s. At $t = 10$ s, minimal overtopping was present indicating that the wave energy dissipated minimally along the slope before reaching the parapet. At $t = 15$ s, the velocity near the structure was higher than Hybrid A, highlighting the direct impact of run-up against the parapet. At $t = 20$ s, the overtopping volume appeared reduced compared with the revetment and Hybrid A configuration, though the local velocity magnitudes at the parapet remained comparable. This behaviour corresponds with the force and overtopping results, where Hybrid B reduced overtopping but experienced substantial parapet loading.

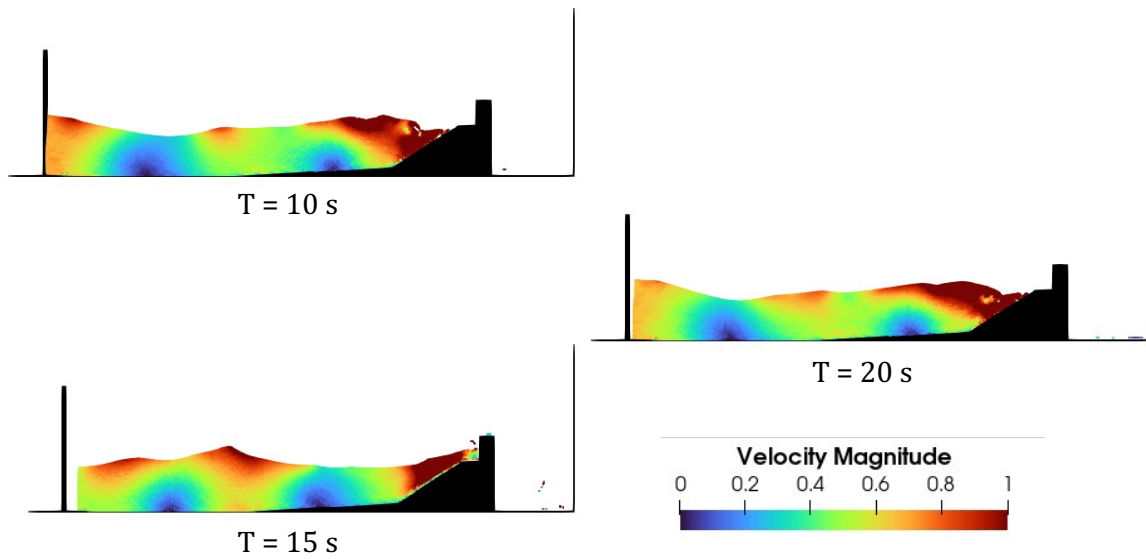


Figure 12: Instantaneous Velocity of Hybrid B (Revetment at SWL)

4.2.4 Optimal Design

The results showed that no single geometry simultaneously minimised both horizontal force and overtopping volume. The revetment was successful in reducing loads but suffered the largest overtopping. The vertical wall had the lowest overtopping among all geometries but experienced higher loading. The hybrid geometries demonstrated enhanced performance by combining both the revetment and vertical wall, though performance varied with crest elevation.

For the hybrid cases, Hybrid A with the revetment crest 0.1 m above the SWL showed that it reduced the peak force on the parapet by 46% compared to Hybrid B. This confirmed that the revetment's role in dissipation wave energy before reaching the vertical wall. However, Hybrid A showcased 42% greater overtopping volume than Hybrid B under T2 conditions. Hybrid B reduced overtopping volumes but experienced higher maximum force with concentrated loads on the parapet.

These findings suggest an effective hybrid should integrate a revetment crest raised above SWL to preserve wave energy dissipation, combined with a taller parapet to meet the overtopping criteria. A pragmatic design would adopt Hybrid A as the baseline with an increased height of the parapet until the overtopping criterion is met.

5 Conclusion

This study employed the SPH framework implemented in DualSPHysics to investigate the hydrodynamic performance of four hard engineering structures, namely a vertical seawall, a smooth impermeable revetment and two hybrid configurations combining a revetment with a parapet of different heights in a two-dimensional wave flume. The methodology was first benchmarked against published SPH and analytical solutions for a vertical wall, with an average error below 2%, thereby instilling confidence in the numerical setup.

The results demonstrated clear performance trade-offs between different coastal defence structures. The vertical seawall experienced one of the lowest overtopping volumes but sustained high horizontal forces due to wave reflection. Conversely, the revetment recorded the lowest horizontal forces owing to its dissipative sloping geometry but yielded the highest overtopping discharge. The hybrid configurations combined the characteristics of both components, where the parapet significantly reduced the overtopping discharge compared to the revetment, but experienced higher horizontal forces. Notably, when the revetment crest was positioned above the still water level (SWL), hydrodynamic forces on the parapet were reduced, emphasising the sensitivity of the hybrid structure's performance to crest elevation.

As a result, the findings suggested that no single structure simultaneously minimised both loading and overtopping. The hybrid arrangement offered the most balanced performance. A practical design approach would adopt a revetment crest raised above SWL to dissipate wave energy, paired with a sufficient tall parapet to contain the wave run-up and maintain overtopping within acceptable limits.

Future research could extend to three-dimensional effects, or alternative geometries such as bullnoses, porous revetments, or hybrid approaches that combine soft engineering measures. These extensions would provide a more comprehensive understanding of an optimal coastal defence design that is resilient under ever-changing climate conditions.

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Appendices

Appendix A: Linear Wave Theory

The formulation of linear wave theory is based on two fundamental conservation laws of fluid mechanics such as conservation of mass and momentum. Under the assumption of the irrotational flow, the fluid velocity can be expressed as the gradient of scalar velocity potential, $\phi(x, z, t)$ such that:

$$\vec{u} = \nabla\phi$$

1

Substituting into the continuity equation yields the Laplace's equation, which governs the behaviour of the velocity potential throughout the fluid domain.

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial z^2} = 0$$

2

The conservation of momentum is represented by the unsteady Bernoulli equation for inviscid and irrotational flow:

$$\frac{\partial\phi}{\partial t} + \frac{p}{\rho} + gz = \text{constant}$$

3

Assuming sinusoidal wave forms and applying the linearised boundary conditions like the kinematic and dynamic condition at the free surface and no-flow condition at the seabed, an analytical solution can be obtained. As both boundary conditions are satisfied simultaneously it leads to the dispersion relation which illustrates the dependence of wave speed on water depth:

$$\omega^2 = gk \tanh(kh)$$

4

where ω is the angular frequency, g is the gravitational acceleration, k is the wave number, and h is the water depth. In deep water ($kh \gg 1$), wave speed depends mainly on wavelength, while in shallow water ($kh \ll 1$), it will be controlled by depth. The

dispersion relationship is vital for understanding wave transformation as oceanic waves from deep water approaches the shore.

Linear wave theory also states that the total mechanical energy per unit width of a wave crest is the sum of potential and kinetic energy. For small-amplitude waves, the energy per unit surface area is given by:

$$E = \frac{1}{8} \rho g H^2$$

5

Where ρ is the water density, g is gravitational acceleration and H is the wave height. The wave crests travel at phase speed $C = \omega/k$, wave energy propagates at group velocity C_g defined as:

$$C_g = \frac{1}{2} C \left(1 + \frac{2kh}{\sinh(2kh)} \right)$$

6

The energy flux, wave power per unit crest width is shown by the product of energy and group velocity as such:

$$P = E \times C_g$$

7